# Moltbook Social Agent
# System Design and Implementation

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## 1 Introduction

The goal of this assignment was to build an agentic AI system that operates in a real online environment—Moltbook. The agent must authenticate, subscribe to `/m/ftec5660`, and then upvote and comment on a given post. Unlike simulated tasks, this requires correct API usage, state awareness, and autonomous decision-making.

## 2 System Architecture

The agent is implemented in Python using LangChain and runs in Google Colab. It consists of the following components:

- **Authentication Module**: Uses an API key stored in Colab secrets to authenticate all requests.
- **Tool Set**: Six tools wrap the Moltbook REST API: `get_feed`, `search_moltbook`, `create_post`, `comment_post`, `upvote_post`, and `subscribe_to_submolt` (added to complete the required tasks).
- **Agent Loop**: A ReAct-style loop that invokes the Gemini model (gemini-2.5-flash) with a system prompt, executes tool calls, and continues until a final answer is produced.
- **Instruction Parser**: The agent interprets a high-level human instruction and breaks it down into sequential tool calls.

A conceptual diagram of the workflow is shown in Figure 1.

**Human Instruction** $\rightarrow$
Gemini model $\rightarrow$ tool call(s) $\rightarrow$ execute API $\rightarrow$ observe result $\rightarrow$ repeat until done
$\downarrow$
Final response

Figure 1: Agent workflow.

## 3 Decision Logic and Autonomy Level

The agent operates with full autonomy once given an instruction. It decides which tools to call, in which order, based on the instruction and the results of previous tool calls. The system prompt enforces rules to prevent spam and ensure professional behaviour.

Key design decisions:

- **Tool completeness**: We added a `subscribe_to_submolt` tool because the provided toolset lacked a way to subscribe. The endpoint was inferred from the Moltbook API documentation (`skill.md`) and confirmed via manual testing.
- **Post ID extraction**: The agent receives the full post URL in the instruction and is capable of extracting the UUID (the last path segment) on its own. This demonstrated the model's ability to handle such patterns without additional helper functions.
- **Sequential execution**: The instruction explicitly lists three tasks; the agent executes them one by one, verifying each step's success before proceeding. The logs show that after a successful subscription, it proceeds to upvote, then comment.
- **Error handling**: The agent logs tool outputs and stops if a critical error occurs (e.g., unauthenticated agent). In our final run, all actions succeeded after the agent was properly claimed.

The autonomy level is high: the agent chooses the tools and their arguments solely from the natural language instruction. No hard-coded steps are used beyond the initial instruction.

## 4 Interaction Logs

Below is the complete log of the agent's execution, showing the successful completion of all required tasks.

```
[13:21:15] [INIT] Starting Moltbook agent loop
[13:21:16] [HUMAN] First, subscribe to the submolt named 'ftec5660'. Then, upvote and
    comment on the post at this URL: https://www.moltbook.com/post/47ff50f3-8255-4dee-87
    f4-2c3637c7351c. Use a professional and friendly comment.
[13:21:16] [TURN] Turn 1/8 started
[13:21:17] [LLM] Model responded
[13:21:17] [LLM.CONTENT] <empty>
[13:21:17] [LLM.TOOL_CALLS] [
  {
    "name": "subscribe",
    "args": { "submolt_name": "ftec5660" },
    "id": "ce49f526-3ab2-44d0-a873-427132b0906c",
    "type": "tool_call"
  }
]
[13:21:17] [TOOL] [1] Calling `subscribe`
[13:21:17] [TOOL.ARGS] { "submolt_name": "ftec5660" }
[13:21:17] [TOOL.RESULT] subscribe finished (success) in 0.39s
[13:21:17] [TOOL.OUTPUT] {
  "success": true,
  "message": "Subscribed to m/ftec5660!   ",
```

```
  "action": "subscribed"
}
[13:21:17] [TURN] Turn 1 completed in 1.7s
[13:21:17] [TURN] Turn 2/8 started
[13:21:18] [LLM] Model responded
[13:21:18] [LLM.CONTENT] <empty>
[13:21:18] [LLM.TOOL_CALLS] [
  {
    "name": "upvote_post",
    "args": { "post_id": "47ff50f3-8255-4dee-87f4-2c3637c7351c" },
    "id": "8df5545c-eb4f-4d97-8934-be4a924ccf15",
    "type": "tool_call"
  }
]
[13:21:18] [TOOL] [1] Calling ‘upvote_post‘
[13:21:18] [TOOL.ARGS] { "post_id": "47ff50f3-8255-4dee-87f4-2c3637c7351c" }
[13:21:19] [TOOL.RESULT] upvote_post finished (success) in 0.17s
[13:21:19] [TOOL.OUTPUT] {
  "success": true,
  "message": "Upvoted!    ",
  "action": "upvoted"
}
[13:21:19] [TURN] Turn 2 completed in 1.22s
[13:21:19] [TURN] Turn 3/8 started
[13:21:20] [LLM] Model responded
[13:21:20] [LLM.CONTENT] <empty>
[13:21:20] [LLM.TOOL_CALLS] [
  {
    "name": "comment_post",
    "args": {
      "post_id": "47ff50f3-8255-4dee-87f4-2c3637c7351c",
      "content": "This is a very insightful post. Thank you for sharing your thoughts!"
    },
    "id": "d10d5f97-ec22-4920-aa3d-cfc25bc25c11",
    "type": "tool_call"
  }
]
[13:21:20] [TOOL] [1] Calling ‘comment_post‘
[13:21:20] [TOOL.ARGS] {
  "post_id": "47ff50f3-8255-4dee-87f4-2c3637c7351c",
  "content": "This is a very insightful post. Thank you for sharing your thoughts!"
}
[13:21:20] [TOOL.RESULT] comment_post finished (success) in 0.2s
[13:21:20] [TOOL.OUTPUT] {
  "success": true,
  "message": "Comment added!   "
}
[13:21:20] [TURN] Turn 3 completed in 1.15s
[13:21:20] [TURN] Turn 4/8 started
[13:21:21] [LLM] Model responded
[13:21:21] [LLM.CONTENT] "I have successfully subscribed to ’ftec5660’, upvoted the post,
     and added a comment."
[13:21:21] [LLM.TOOL_CALLS] []
[13:21:21] [STOP] No tool calls  final answer produced in 0.88s
```

```
Final agent response: I have successfully subscribed to 'ftec5660', upvoted the post, and
     added a comment.
```

The logs confirm that the agent correctly interpreted the instruction, subscribed to the submolt, upvoted the post, and left a professional comment. No manual intervention was required after the initial instruction.

# 5 Conclusion

The Moltbook social agent successfully completed all assigned tasks autonomously. By extending the toolset and providing a clear instruction, the agent demonstrated the ability to navigate a real online platform, subscribe to a community, and interact with content. The design is modular and can be adapted to similar social APIs.